

# A Proposal For Scrutineering BLDC Motors and Electronic Commutators (eCom)

I would like to recommend a way to accurately measure configurations of brushless motors and eComs by measuring motor speed at different voltages, and offer to provide a prototype measuring device to the BSCRA to test the recommended method.

This offer is a result of the recent BSCRA proposal to permit brushless motors in the intro class which has highlighted the need to be able to compare a specific motor/eCom combination as part of the scrutineering process.

## The Measurement Device

The measurement device shall be a tachometer, comprised of: a small microprocessor-driven device with an LCD display fed by a Hall Effect probe.

The Hall Effect probe would detect the external magnetic field from the motor's rotor without having to have any physical contact with the motor or a need for reflective tape if for example an optical tachometer was used.

The software in the device would sample the motor speed at different voltages, output values to the LCD (and possibly other destinations), and provide the motor's voltage constant – which is proved below to be an accurate measurement for comparison.

The specification and design of the device shall be open source so that its operation can be examined and copied by anyone else would like to make their own.

## Summary

The important parameters that need to be measured to compare one motor/eCom configuration to another, are:

- Motor speed.
- Motor power.
- eCom setup.
- Motor mechanical dimensions

Speed can be measured with a tachometer. Power can be derived from the speed measurement. eComs can be compared by measuring speed at fixed voltages. The dimensions can be checked by using a vernier calliper.

## Power Measurement

In mechanical terms, power is expressed as:

$$\text{power} = \text{speed} \times \text{torque}$$

Torque is directly proportional to current, which is expressed as:

$$T = K_t \times I \text{ (where } K_t \text{ is the motor torque constant, and } I \text{ is the current)}$$

Ohm's law ( $V=IR$ ) says the maximum current is limited by the total circuit resistance which is the sum of resistance in: the track wiring, fuses, plugs and sockets, the controller, the track braid, track braid to car braid, eCom resistance and motor resistance.

Differences in eCom resistance and motor resistance from one unit to another will be small compared to all the other resistance sources. Therefore the maximum current, and hence the maximum torque and power, are independent of the motor and eCom, if the motor  $K_t$  values are similar.

Therefore the  $K_t$  value is good a measurement of power, and as shown below can be derived from a speed measurement.

## Ecom Measurement

To ensure the eCom has standard firmware, scrutineering can be accomplished for a motor of a fixed mechanical size by measuring the speed at a fixed supply voltage, or more rigorously at two or three fixed voltages say 3V, 7V or 8V and 12V.

Some variation about nominal values would have to be determined to allow for manufacturing tolerances. Any motor, eCom combination that met the size and speed versus voltage specification could be allowed.

## Background

To compare motors we need to compare their voltage constant-  $K_e$ . this is different from  $K_v$ .  $K_v$  is a convenient measure of a motor for hobby use, and its units are rpm/volt.  $K_v$  is often called a voltage constant which is really a misnomer as it's actually a speed constant, or the inverse of a true voltage constant. A more accurate description of  $K_v$  is a velocity constant as it is the rotational velocity in rpm to produce a motor back emf of 1V- for an unloaded motor the speed is approximately  $K_v \times$  applied voltage (rpm).

### Proof

Using the correct scientific units and with a few calculations it can be shown that only the BLDC motor and eCom voltage constant needs to be determined.

For SI units, see: <https://www.bipm.org/en/measurement-units/>

#### Definition of symbols

|                                                 |          |               |
|-------------------------------------------------|----------|---------------|
| Back emf constant (not $K_v$ but related to it) | $K_e$    | V/(radians/s) |
| Torque constant                                 | $K_t$    | Nm/A          |
| Motor shaft angular velocity                    | $\omega$ | radians/s     |
| Back emf                                        | $V_e$    | V             |
| Current to eCom                                 | $I$      | A             |
| Torque                                          | $T$      | Nm            |
| Electrical power converted to mechanical power  | $P_e$    | W             |
| Motor mechanical power                          | $P_m$    | W             |

#### Conversion of common units to SI units

|                              |                                                |               |
|------------------------------|------------------------------------------------|---------------|
| Rotation to angle            | 1 revolution = $360^\circ = 2 \times \pi$      | radians       |
| Motor shaft angular velocity | $\omega = (\text{rpm}/60) \times 2 \times \pi$ | radians/s     |
| Back emf constant            | $K_e = (1/K_v) \times 60/2\pi$                 | V/(radians/s) |

#### Equations

|                                                        |                                    |    |
|--------------------------------------------------------|------------------------------------|----|
| Back emf                                               | $V_e = K_e \times \omega$          | V  |
| Torque                                                 | $T = K_t \times I$                 | Nm |
| Electrical power that is converted to mechanical power | $P_e = V_e \times I$               | W  |
| Substitute for $V_e$                                   | $P_e = K_e \times \omega \times I$ | W  |
| Mechanical power                                       | $P_m = \omega \times T$            | W  |
| Substitute for $T$                                     | $P_m = \omega \times K_t \times I$ | W  |

As the electrical power converted to mechanical power must equal the mechanical power, then we can calculate  $K_t$  from  $K_e$

|                                              |                                                           |                       |
|----------------------------------------------|-----------------------------------------------------------|-----------------------|
| Electrical power equals the mechanical power | $P_e = P_m$                                               | W                     |
| Substitute for $P_e$ and $P_m$               | $K_e \times \omega \times I = \omega \times K_t \times I$ | W                     |
| Eliminate $\omega \times I$                  | <u><math>K_e = K_t</math></u>                             | V/(radians/s) or Nm/A |

So if the (voltage constant),  $K_e$  is measured then (torque constant)  $K_t$  is also known.

A close approximation to  $K_e$  can be determined by applying a known voltage to the eCom and measuring the rotational velocity of the unloaded motor. The losses in the motor and eCom mean the actual voltage supplied will be a little higher than the back emf. As it is a comparative measurement between different motors and eComs it should be of no consequence.

The power a motor can produce is determined by how much current can be forced into the motor while keeping it cool. Physically larger motors and more powerful eComs are able to cope with higher current and therefore more power. For the same  $K_e$  and size of motor the wire gauge and number of turns is very similar so the resistance will be very similar. Other losses in the motor are also closely matched, therefore their current rating will be almost identical.